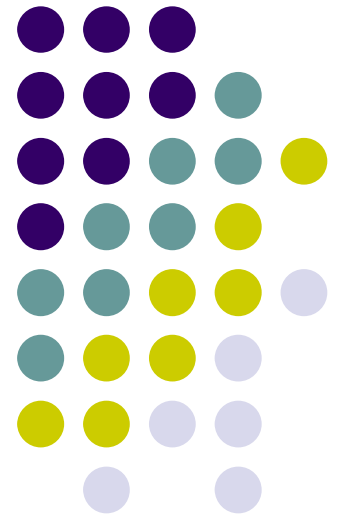
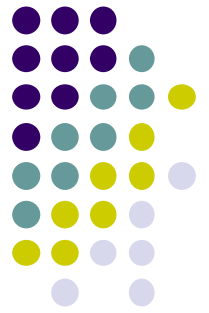


WIA1005 Network Technology Foundation

Chapter 2

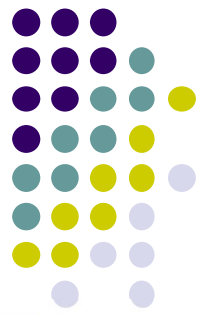
Network Protocols and Network Operating Systems





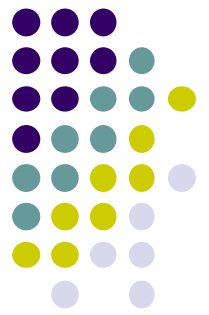
Contents

- Rules of Communication
- Network Protocols
- Data Encapsulation
- Data Access
- Network Operating Systems
- Basic Command



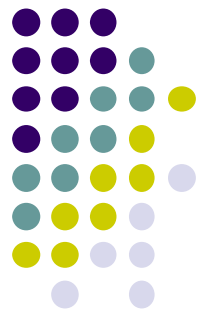
Rules of Communication

- Communication begins with a **message**, or **information**, that must be sent from a **source** to a **destination**. The sending of this message, whether by face-to-face communication or over a network, is governed by rules called protocols.
- The **protocols** used are specific to the characteristics of the communication method, including the characteristics of the **source, destination and channel**.
- Protocol Requirements
 - An identified **sender** and **receiver**
 - Common language and grammar
 - **Speed** and **timing** of delivery
 - Confirmation or acknowledgment requirements



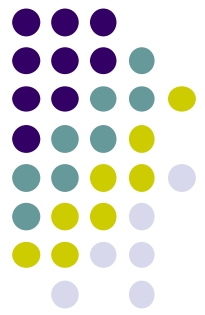
Rules of Communication

- Common computer protocols
 - message encoding
 - message formatting and encapsulation
 - message size
 - message timing
 - message delivery options
- **Encoding** is the process of converting information into another, acceptable form, for transmission. **Decoding** reverses this process in order to interpret the information.
- When a message is sent from source to destination, it must use a specific format or structure. Message formats depend on the type of message and the channel that is used to deliver the message.



Rules of Communication

- When communication happens, the messages usually broken into smaller parts or sentences. These sentences are limited in size to what the receiver can process at one time.
- Message timing includes the following:
 - **Flow Control** - Flow control defines how much information can be sent and the speed at which it can be delivered.
 - **Response Timeout** - Hosts on the network use network protocols that specify how long to wait for responses and what action to take if a response timeout occurs.
 - **Access method** - This determines when someone can send a message.

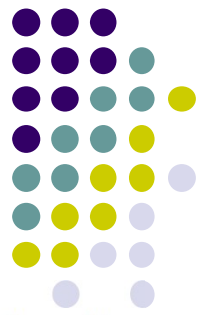


Rules of Communication

- A message may need to be best delivered in different ways. Sometimes, a person wants to communicate information to a single individual. At other times, the person may need to send information to a group of people at the same time, or even to all people in the same area.

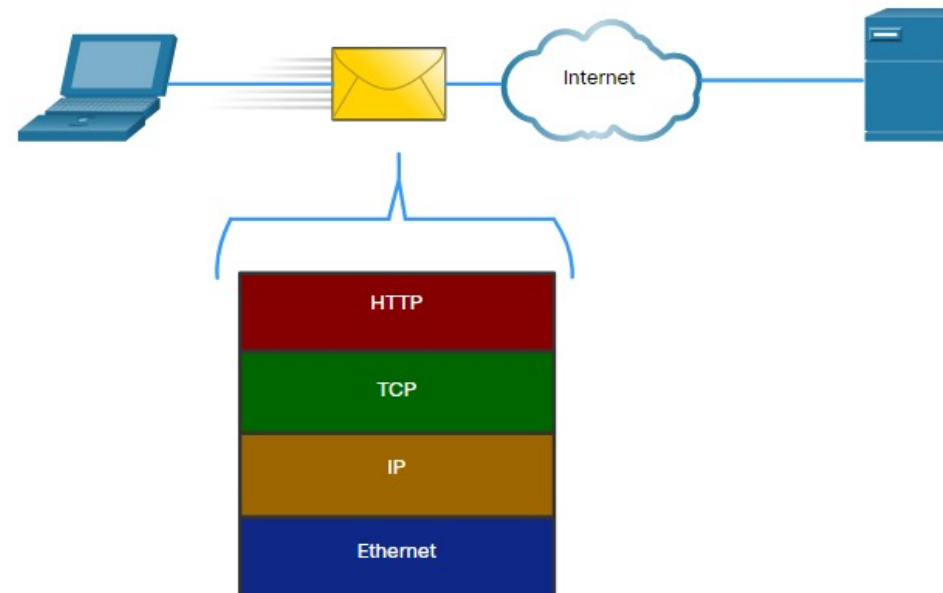
Computer Communication

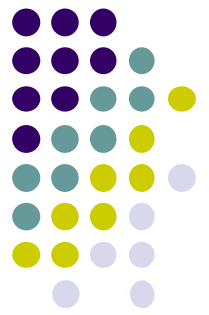




Network Protocols

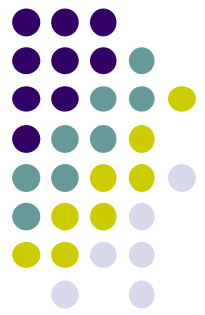
- Networking protocols define a common format and set of rules for exchanging messages between devices.
- A message sent over a computer network typically requires the use of several protocols, each one with its own functions and format.



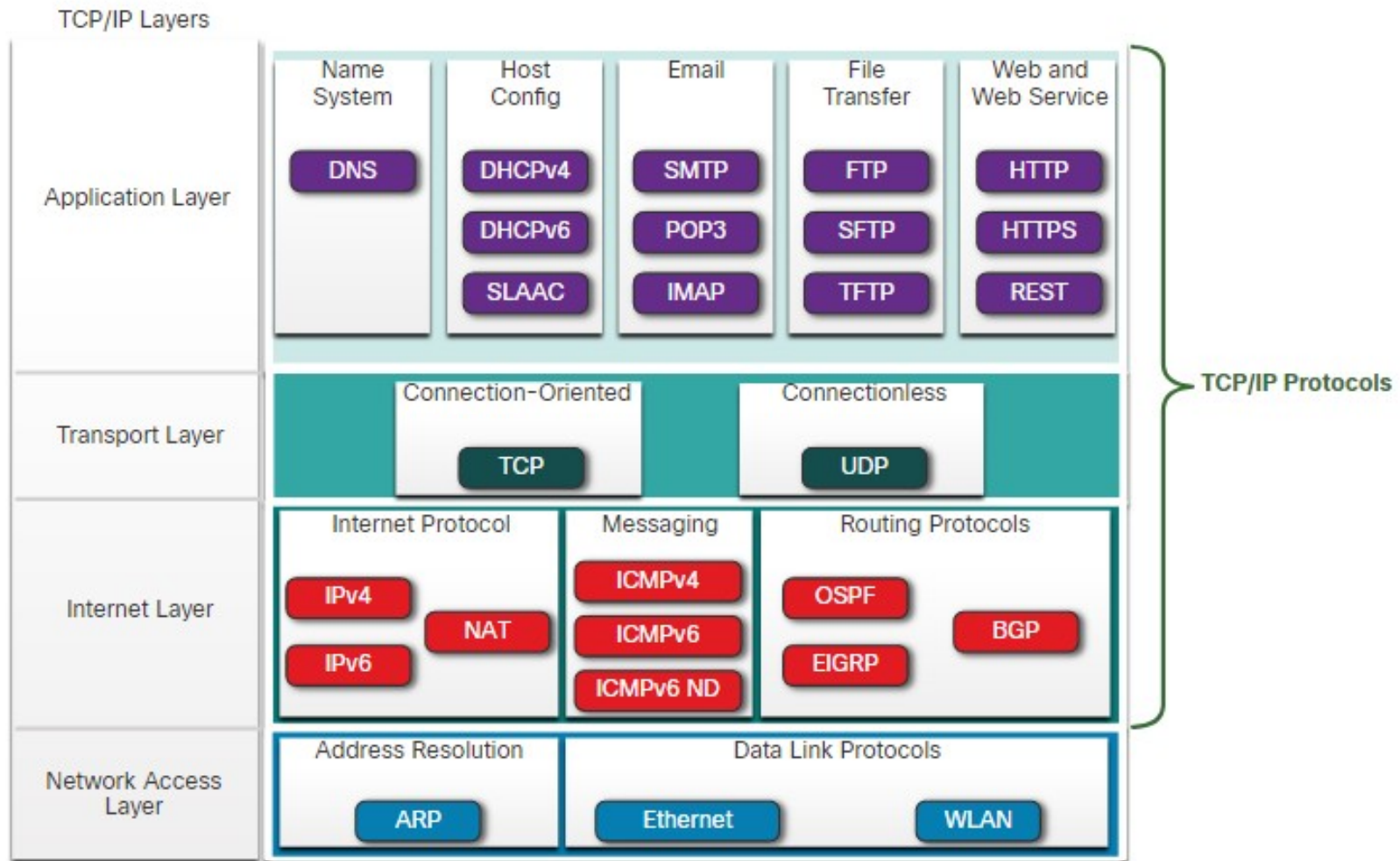


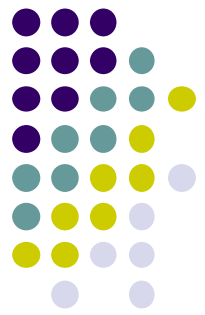
Network Protocols

- A group of **inter-related protocols** necessary to perform a communication function is called a **protocol suite**.
- Protocol suites are implemented by **hosts and networking** devices in software, hardware or both.
- The **TCP/IP suite** is a suite of protocols required for **transmitting and receiving** information using the Internet.
- A layered model, such as the TCP/IP model, is often used to help visualize the interaction between various protocols. A layered model depicts the operation of the protocols occurring within each layer, as well as the interaction of protocols with the layers above and below each layer.



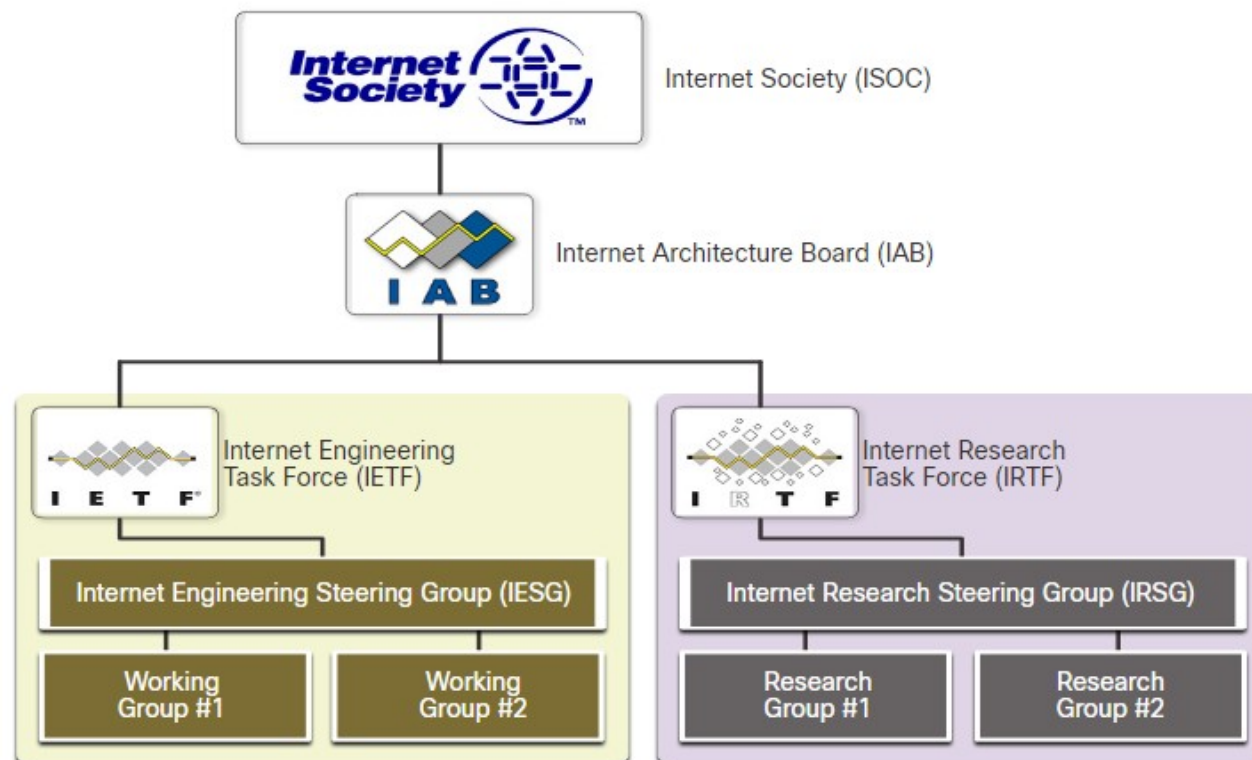
Network Protocols

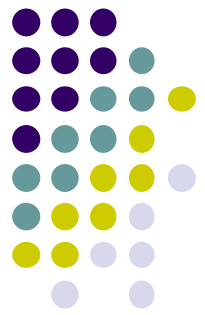




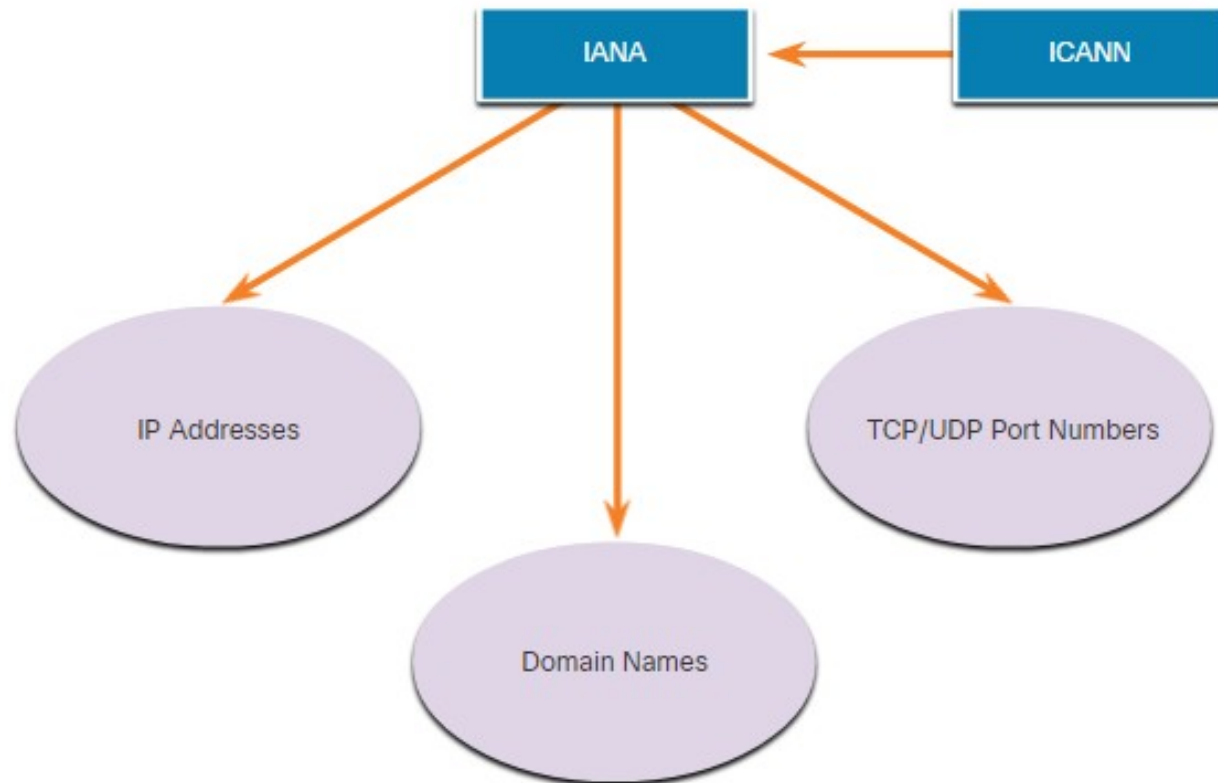
Network Protocols

- Standards organizations are usually vendor-neutral, non-profit organizations established to develop and promote the concept of open standards.

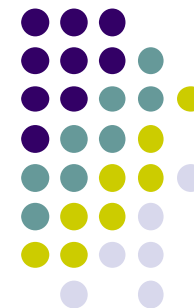




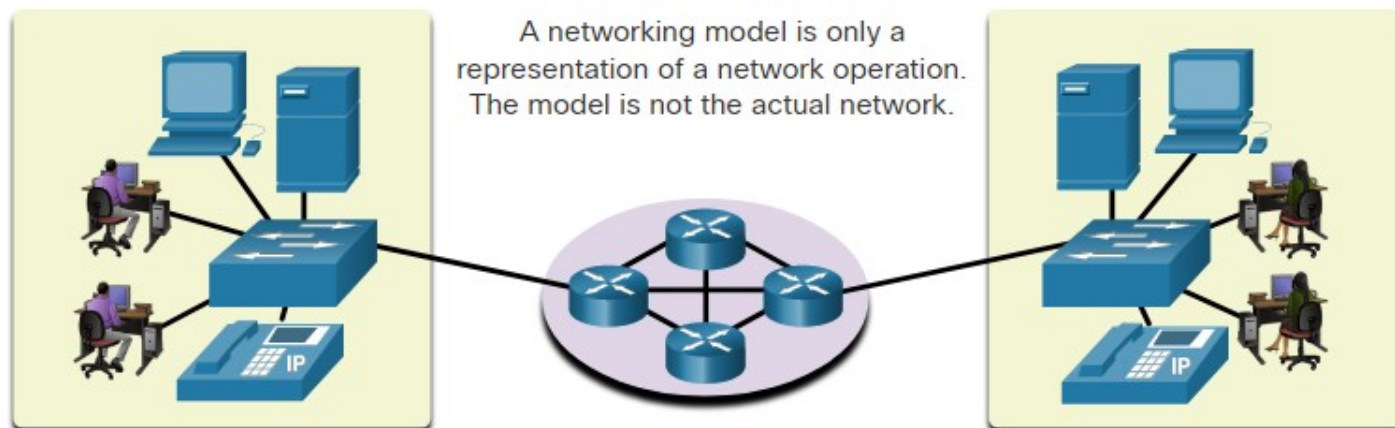
Network Protocols



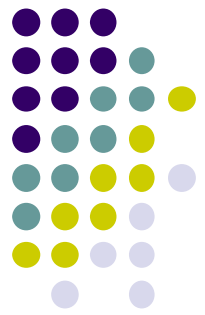
- **Internet Corporation for Assigned Names and Numbers (ICANN)** - Based in the United States, ICANN coordinates IP address allocation, the management of domain names, and assignment of other information used in TCP/IP protocols.
- **Internet Assigned Numbers Authority (IANA)** - Responsible for overseeing and managing IP address allocation, domain name management, and protocol identifiers for ICANN.



Network Protocols

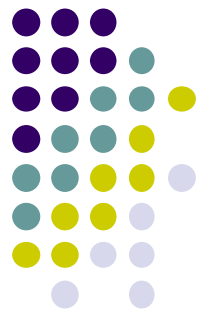


OSI Model	TCP/IP Protocol Suite	TCP/IP Model
Application	HTTP, DNS, DHCP, FTP	Application
Presentation		
Session		
Transport	TCP, UDP	Transport
Network	IPv4, IPv6, ICMPv4, ICMPv6	Internet
Data Link	Ethernet, WLAN, SONET, SDH	Network Access
Physical		



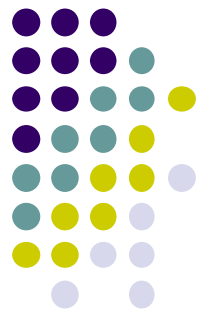
Network Protocols

OSI Model Layer	Description
7 - Application	The application layer contains protocols used for process-to-process communications.
6 - Presentation	The presentation layer provides for common representation of the data transferred between application layer services.
5 - Session	The session layer provides services to the presentation layer to organize its dialogue and to manage data exchange.
4 - Transport	The transport layer defines services to segment, transfer, and reassemble the data for individual communications between the end devices.
3 - Network	The network layer provides services to exchange the individual pieces of data over the network between identified end devices.
2 - Data Link	The data link layer protocols describe methods for exchanging data frames between devices over a common media
1 - Physical	The physical layer protocols describe the mechanical, electrical, functional, and procedural means to activate, maintain, and de-activate physical connections for a bit transmission to and from a network device.



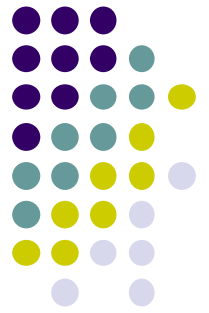
Network Protocols

TCP/IP Model Layer	Description
4 - Application	Represents data to the user, plus encoding and dialog control.
3 - Transport	Supports communication between various devices across diverse networks.
2 - Internet	Determines the best path through the network.
1 - Network Access	Controls the hardware devices and media that make up the network.



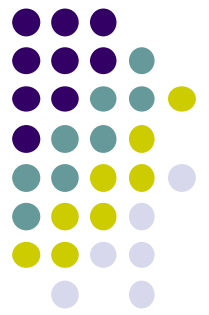
Data Encapsulation

- A single communication, such as a video or an email message with many large attachments, could be sent across a network from a source to a destination as one massive, uninterrupted stream of bits.
- However, this would create problems for other devices needing to use the same communication channels or links. These large streams of data would result in significant delays.
- **Segmentation** is the process to divide the data into smaller, more manageable pieces to send over the network. Segmentation can increase the reliability of network communications.



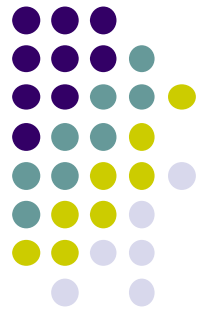
Data Encapsulation

- By sending smaller individual pieces from source to destination, many different conversations can be interleaved on the network.
- The process used to interleave the pieces of separate conversations together on the network is called **multiplexing**.
- Each segment of the message must go through a similar process to ensure that it gets to the correct destination and can be reassembled into the content of the original message.

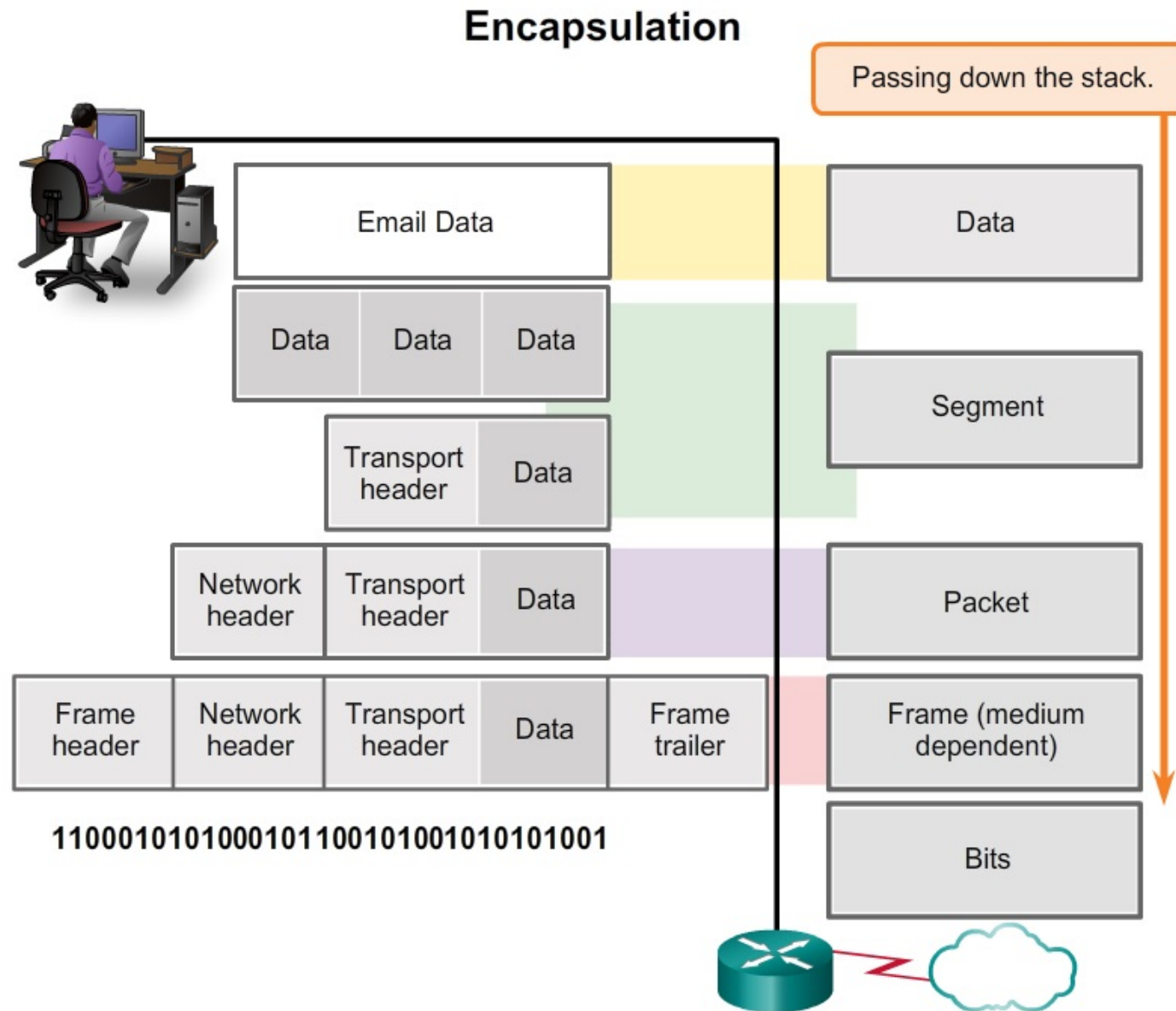


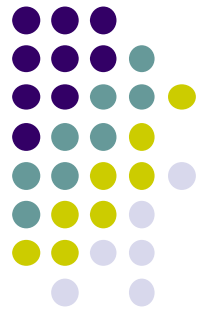
Data Encapsulation

- As application data is passed down the protocol stack on its way to be transmitted across the network media, various protocols add information to it at each level. This is commonly known as the **encapsulation** process.
- The form that a piece of data takes at any layer is called a **protocol data unit (PDU)**.
 - **Data** – Application layer PDU
 - **Segment** - Transport layer PDU
 - **Packet** - Internet layer PDU
 - **Frame** - Network access layer PDU
 - **Bits** – Physical layer PDU



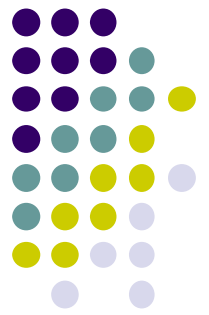
Data Encapsulation





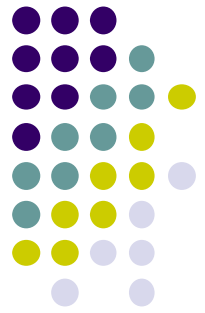
Data Encapsulation

- This process is reversed at the receiving host, and is known as de-encapsulation. **De-encapsulation** is the process used by a receiving device to remove one or more of the protocol headers.
- The data is de-encapsulated as it moves up the stack toward the end-user application.

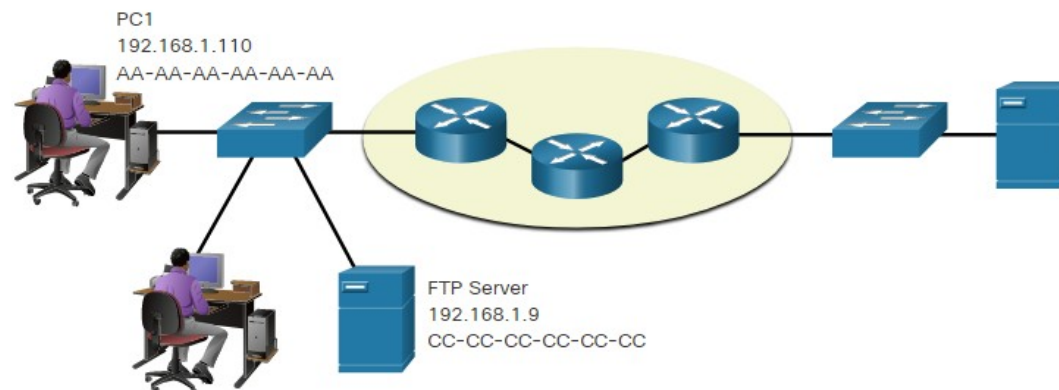
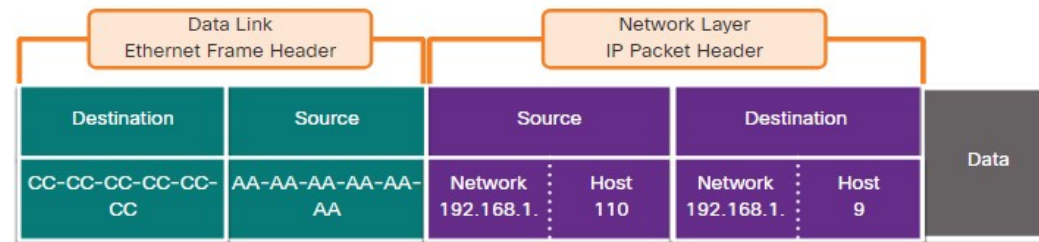


Data Access

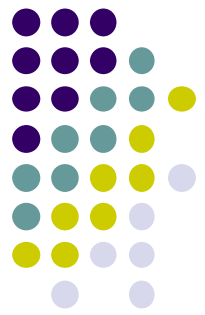
- The network and data link layers are responsible for delivering the data from the source device to the destination device using different addresses:
- **Network layer source and destination addresses** - Responsible for delivering the IP packet from the original source to the final destination, which may be on the same network or a remote network.
- **Data link layer source and destination addresses** - Responsible for delivering the data link frame from one network interface card (NIC) to another NIC on the same network.



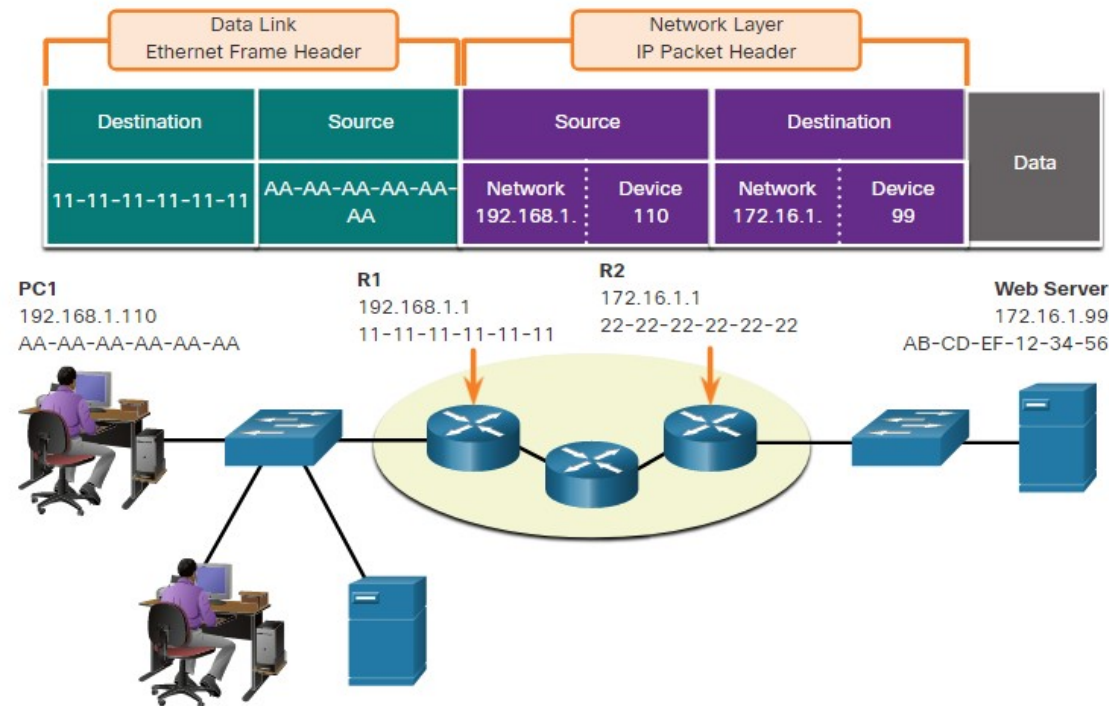
Data Access



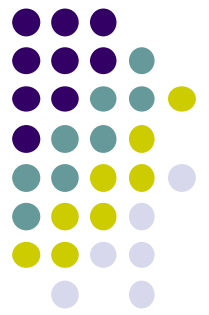
In the same network, the data link frame is sent directly to the receiving device. The data link addresses are known as Media Access Control (MAC) addresses in Ethernet



Data Access



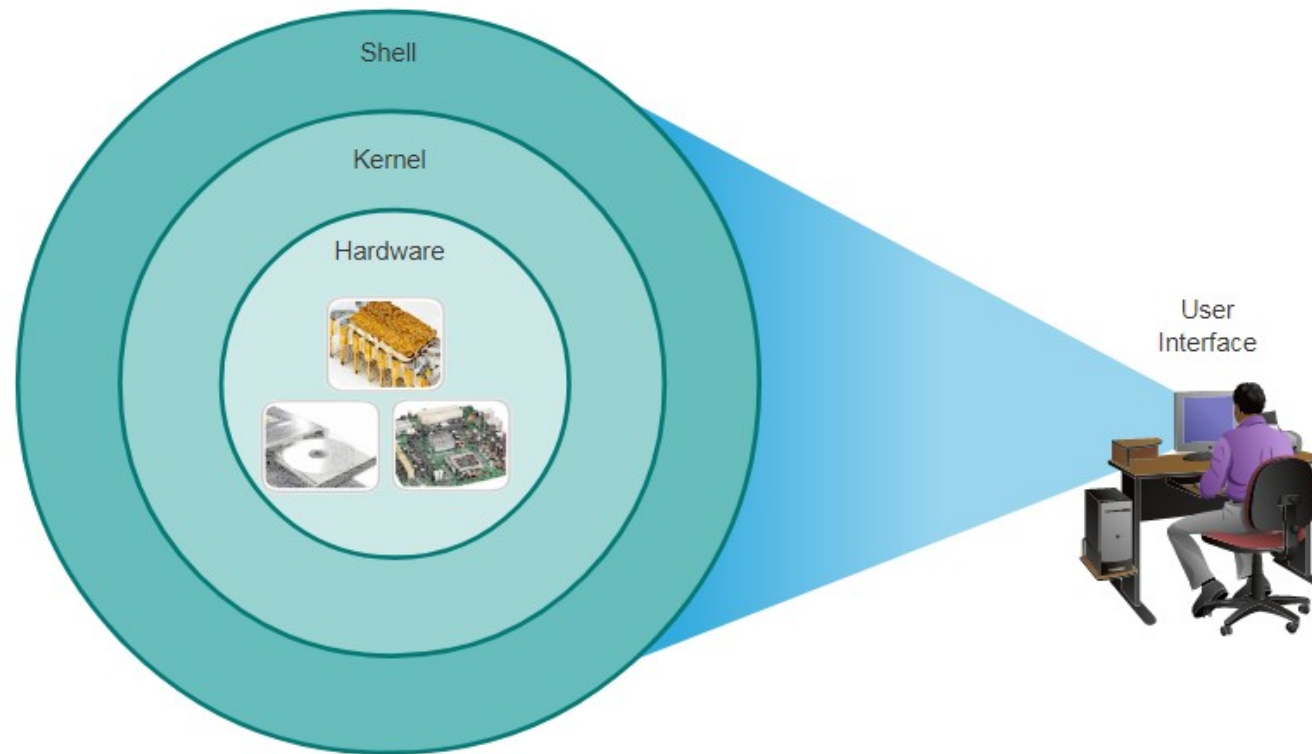
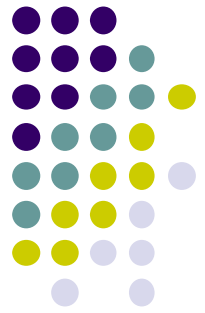
When the sender of the packet is on a different network from the receiver, the source and destination IP addresses will represent hosts on different networks. The frame must be sent to another the router or default gateway.

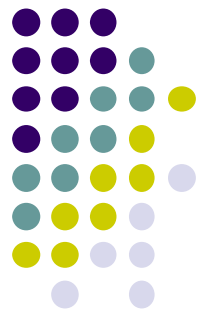


Network Operating Systems

- Infrastructure network devices use a network operating system. Network operating systems are in many ways similar to the operating systems of PCs.
- The network operating system used on Cisco devices is called the Cisco **Internetwork Operating System (IOS)**.
- The IOS file itself is stored in a semi-permanent memory area called flash that provides non-volatile storage.
- The IOS is copied from flash into random access memory (RAM) when the device is powered on. The IOS then runs from RAM when the device is operating.
- Kernel interacts directly with computer hardware. User can interact with the shell using either the command-line interface (CLI) or graphical user interface (GUI).

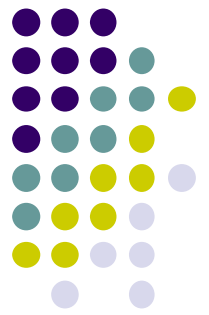
Network Operating Systems





Network Operating Systems

- There are several ways to access the CLI environment.
- **Console**
 - The console port is a management port that provides out-of-band access. **Out-of-band access** refers to access via a dedicated management channel that is used for device maintenance purposes only.
- **AUX**
 - An older way to establish a CLI session remotely is via a telephone dialup connection using a modem connected to the auxiliary (AUX) port of a router. The AUX method is also an **out-of-band** connection and does not require any networking services to be configured or available on the device.



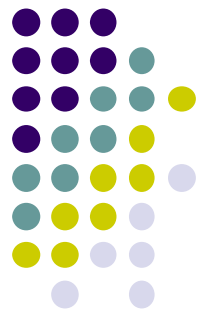
Network Operating Systems

- **Telnet**

- Telnet is a method for remotely establishing a CLI session of a device, through a virtual interface, over a network. Unlike the console connection, Telnet sessions require active networking services on the device.

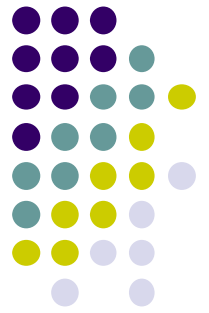
- **SSH**

- The Secure Shell (SSH) protocol provides a remote login similar to Telnet, except that it uses more secure network services. SSH provides stronger password authentication than Telnet and uses encryption when transporting session data.



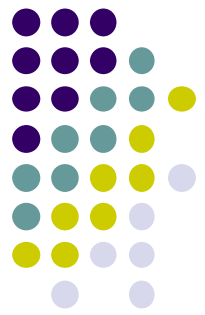
Network Operating Systems

- There are a number of terminal emulation programs to connect to a networking device either by a serial connection over a console port or by a Telnet/SSH connection.
 - PuTTY
 - Tera Term
 - SecureCRT
 - HyperTerminal
 - OS X Terminal



Network Operating System

- The CLI uses a hierarchical structure for the modes. The major modes are:
 - User executive (User EXEC) mode
 - Privileged executive (Privileged EXEC) mode
 - Global configuration mode
 - Other specific configuration modes, such as interface configuration mode or line configuration mode



Network Operating System

User EXEC Command-Router>

ping
show (limited)
enable
etc.

Privileged EXEC Commands-Router#

all User EXEC commands
debug commands
reload
configure
etc.

Global Configuration Commands-Router(config)#

hostname
enable secret
ip route

interface ethernet
serial
dsl
etc.

router rip
ospf
eigrp
etc.

line vty
console
etc.

Interface Commands-Router(config-if)#

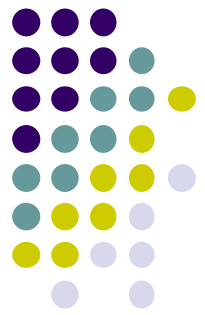
ip address
ipv6 address
encapsulation
shutdown/no shutdown
etc.

Routing Engine Commands-Router(config-router)#

network
version
auto summary
etc.

Line Commands-Router(config-line)#

password
login
modem commands
etc.



Network Operating System

User EXEC Mode

Limited examination of router.
Remote access.

```
Switch>  
Router>
```

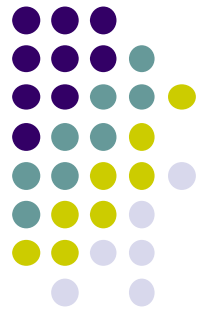
The **User EXEC** mode allows only a limited number of basic monitoring commands and is often referred to as view-only mode.

Privileged EXEC Mode

Detailed examination of router. Debugging and testing. File manipulation. Remote access.

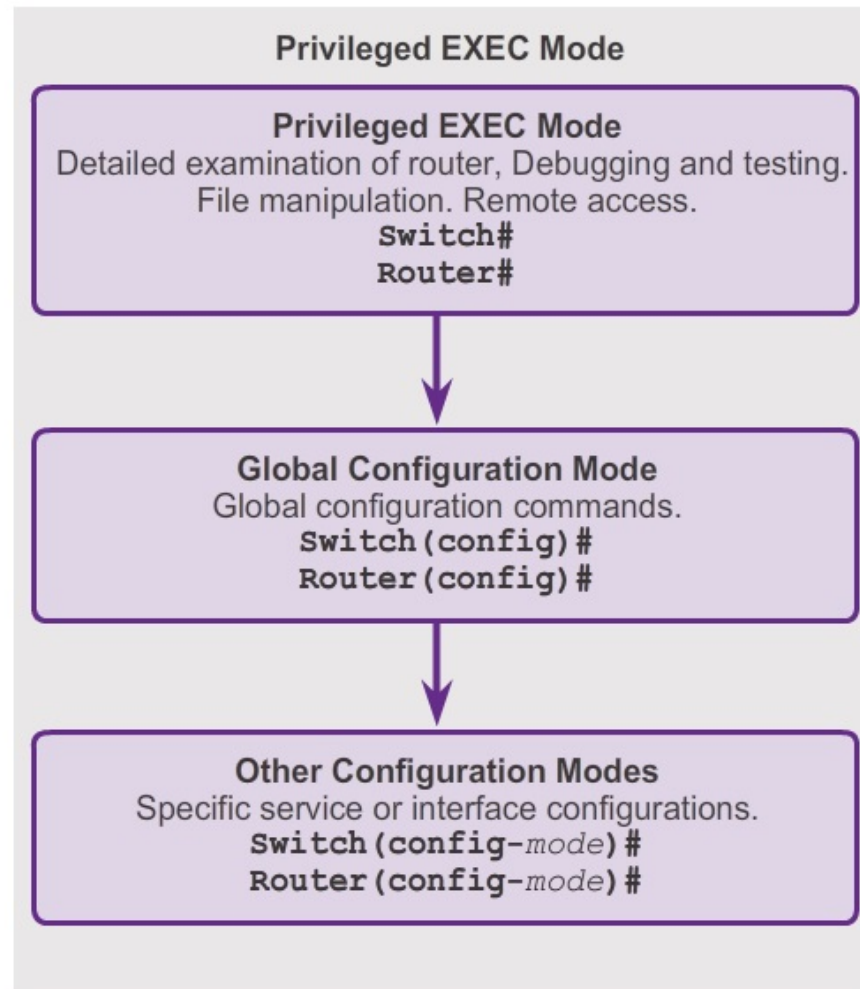
```
Switch#  
Router#
```

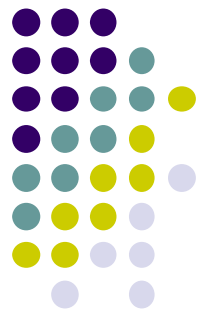
The **Privileged EXEC** mode, by default, allows all monitoring commands, as well as execution of configuration and management commands.



Network Operating System

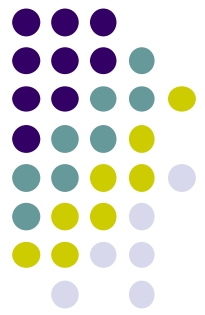
Within Privileged EXEC mode, network administrators can access the global configuration mode and all other sub-configuration modes.





Basic Command

- Hostnames allow devices to be identified by network administrators over a network or the Internet.
- Passwords are the primary defense against unauthorized access to network devices.
 - **Enable password** - Limits access to the privileged EXEC mode
 - **Enable secret** - Encrypted, limits access to the privileged EXEC mode
 - **Console password** - Limits device access using the console connection
 - **VTY password** - Limits device access over Telnet



Basic Command

Configure the switch hostname to be 'Sw-Floor-1'.

```
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
```

Sw-Floor-1#**conf terminal**

Sw-Floor-1(config)#**enable secret class**

Sw-Floor-1(config)#**line console 0**

Sw-Floor-1(config-line)#**password cisco**

Sw-Floor-1(config-line)#**login**

Sw-Floor-1(config-line)#**exit**

Sw-Floor-1(config)#

Sw-Floor-1(config)#**line vty 0 15**

Sw-Floor-1(config-line)#**password cisco**

Sw-Floor-1(config-line)#**login**

Enter the command to encrypt the plain text passwords.

```
Switch(config)# service password-encryption
```

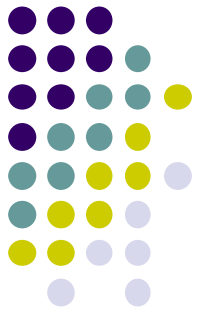
Exit global configuration mode and view the running configuration.

```
Switch(config)# exit
```

```
Switch# show running-config
```

Sw-Floor-1#**disable**

Sw-Floor-1>**enable**



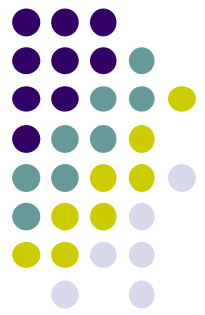
Basic Command

```
Sw1-Floor-1(config)#banner motd # This is a secure system. Authorized Access ONLY!!! #
```

This configuration results in this message of the day banner.

Delimiting characters are not included in the message.

```
Sw1-Floor-1 con0 is now available
Press RETURN to get started.
This is a secure system. Authorized
Access ONLY!!!
User Access Verification
password:
Sw1-Floor-1>enable
Password:
Sw1-Floor-1#
```

Basic Command

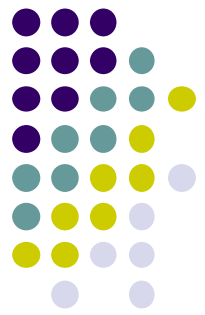
```
Switch#show running-config
```

Lists the complete configuration currently active in RAM.

```
Switch#show running-config  
Building configuration...  
Current configuration : 2904 bytes  
!  
! Last configuration change at 00:02:32  
UTC Mon Mar 1 1993  
!  
version 15.0  
no service pad  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
!  
<output omitted>  
!
```

The active configuration can be copied to NVRAM.

```
Switch#copy running-config startup-config
```



Basic Command

Configuring a Switch Virtual Interface

Enter interface configuration mode for VLAN 1.

```
Switch(config)# interface vlan 1
```

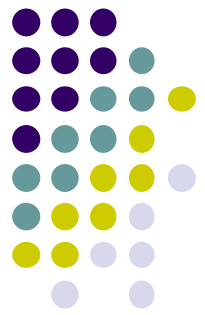
Configure the IP address as '192.168.10.2' and the subnet mask as '255.255.255.0'.

```
Switch(config-if)# ip address 192.168.10.2 255.255.255.0
```

Activate the interface.

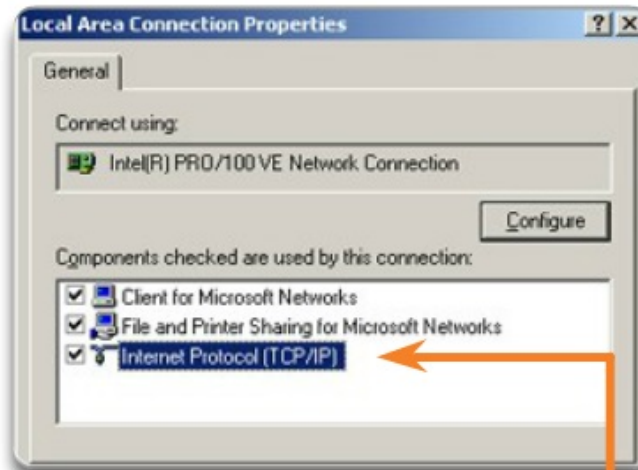
```
Switch(config-if)# no shutdown
```

```
%LINK-5-CHANGED: Interface Vlan1, changed state to up
```



Basic Command

Addressing End Devices



For manual static assignments, enter addresses:

IP address
Subnet mask
Default gateway

